



A-Level Economics

7136/3 Paper 3 Economic Principles And Issues
Predicted Paper Final Mark scheme

7136
Predicted Paper

Version/Stage: v1.0 [Set C — Advanced]

Mark schemes are prepared by the Lead Assessment Writer and considered, together with the relevant questions, by a panel of subject teachers. This mark scheme includes any amendments made at the standardisation events which all associates participate in and is the scheme which was used by them in this examination. The standardisation process ensures that the mark scheme covers the students' responses to questions and that every associate understands and applies it in the same correct way. As preparation for standardisation each associate analyses a number of students' scripts. Alternative answers not already covered by the mark scheme are discussed and legislated for. If, after the standardisation process, associates encounter unusual answers which have not been raised they are required to refer these to the Lead Assessment Writer.

It must be stressed that a mark scheme is a working document, in many cases further developed and expanded on the basis of students' reactions to a particular paper. Assumptions about future mark schemes on the basis of one year's document should be avoided; whilst the guiding principles of assessment remain constant, details will change, depending on the content of a particular examination paper.

Further copies of this mark scheme are available from aqa.org.uk

This is a predicted paper intended for revision purposes only. It is not an official AQA publication. The mark scheme follows the style and structure of AQA published materials to support realistic practice.

Annotations

Annotation	Description
?	Unclear
AN	Analysis
APP	Application
BOD	Benefit of the doubt
EVAL	Evaluation
Highlight	Highlight
IR	Irrelevant
KU	Knowledge and understanding
NAQ	Not answered question
REP	Repetition
SEEN	Indicates that the point has been noted, but no credit has been given.
Tick	Correct point
Cross	Incorrect point

Level of response marking instructions

Level of response mark schemes are broken down into levels, each of which has a descriptor. The descriptor for the level shows the average performance for the level. There are marks in each level.

Before you apply the mark scheme to a student's answer read through the answer and annotate it (as instructed) to show the qualities that are being looked for. You can then apply the mark scheme.

Step 1 — Determine a level

Start at the lowest level of the mark scheme and use it as a ladder to see whether the answer meets the descriptor for that level. The descriptor for the level indicates the different qualities that might be seen in the student's answer for that level. If it meets the lowest level then go to the next one and decide if it meets this level, and so on, until you have a match between the level descriptor and the answer. With practice and familiarity you will find that for better answers you will be able to quickly skip through the lower levels of the mark scheme.

When assigning a level you should look at the overall quality of the answer and not look to pick holes in small and specific parts of the answer where the student has not performed quite as well as the rest. If the answer covers different aspects of different levels of the mark scheme you should use a best-fit approach for defining the level and then use the variability of the response to help decide the mark within the level.

Step 2 — Determine a mark

Once you have assigned a level you need to decide on the mark. The descriptors on how to allocate marks can help with this. You may well need to read back through the answer as you apply the mark scheme to clarify points and assure yourself that the level and the mark are appropriate.

Indicative content in the mark scheme is provided as a guide for examiners. It is not intended to be exhaustive and you must credit other valid points. Students do not have to cover all of the points mentioned in the indicative content to reach the highest level of the mark scheme.

An answer which contains nothing of relevance to the question must be awarded no marks.

Section A — Multiple Choice Answers

Each question is worth 1 mark. The correct answers and brief justifications are shown in the grid below.

Question	Answer	Brief explanation
01	B	Two-sided platforms often price one side below MC (even free) to attract users who generate indirect network benefits for the paying side.
02	A	Hysteresis: temporary demand shocks leave permanent scars (skills loss, disconnection from labour market), raising the natural rate.
03	C	First-degree / perfect price discrimination extracts all consumer surplus BUT produces at the competitive output (no deadweight loss).
04	B	Coase theorem: with well-defined property rights and low transaction costs, private bargaining can reach the efficient outcome without government action.
05	C	In the Solow model, diminishing returns to capital mean that long-run growth in output per worker comes from exogenous technological progress.
06	C	Lump-sum tax = fixed amount regardless of behaviour → does not distort economic decisions at the margin, hence minimal deadweight loss.
07	B	Balance of payments identity: $CA + KA = 0$ (in a pure floating system). Capital-importer has KA surplus matched by CA deficit.
08	B	Adverse selection: when insurers can't distinguish risks, the highest-risk customers buy disproportionately, driving up premiums.
09	C	Natural rate hypothesis: LRPC is vertical at the NRU; no long-run trade-off.
10	C	Taylor rule: 1pp above target → raise rate by MORE than 1pp to raise real interest rate and tighten.
11	D	$P = £25$ throughout. MC per unit: £8 ($Q=10 \rightarrow 20$), £9 ($Q=20 \rightarrow 30$), £11 ($Q=30 \rightarrow 40$), £14 ($Q=40 \rightarrow 50$). $P > MC$ at every level shown. Profit at $Q=50$: $TR = £1,250$, $TC = £580$, profit = £670 (the highest in the table). Firm produces $Q = 50$.
12	B	Classic prisoners' dilemma. Each firm's dominant strategy is Low price ($14 > 12$ if opponent High; $6 > 4$ if opponent Low). Nash equilibrium: (Low, Low) with payoff (6, 6) — both worse off than (High, High).
13	A	Nominal wage rose 23.3% ($18.50/15.00$). CPI rose 23.6% ($123.6/100$). Real wage change = $+23.3\% - 23.6\% \approx -0.3\%$. Narrow but real wage fell.
14	C	Sterling depreciated 16% ($ERI\ 100 \rightarrow 83.6$), which would normally boost competitiveness. But real unit labour costs rose by 1.8% — eroding the competitiveness gain. Net: improvement but less than the nominal depreciation suggests.
15	B	AI-assisted worker produces 2x output at the same hours. Firms will substitute toward AI-assisted workers, reducing demand for non-AI-assisted labour.
16	C	Gap = MEC – current carbon price. Power: £30; Heavy industry: £35; Aviation: £65; Agriculture: £60. Aviation has the largest gap between social cost and price signal.
17	C	Reduction in Gini: A: 0.15. B: 0.08. C: 0.26. D: 0.06. C has the largest reduction → strongest redistribution.
18	D	Tariff revenue = tariff rate × quantity imported. Imports at £13 = demand – supply = $85 - 35 = 50$. Revenue = $50 \times £3 = £150$. Answer D states this calculation correctly.
19	B	Top 10% own 43%. Bottom 50% (deciles 2–5) own 9% + roughly 0% = 9%. $43\% > 9\%$ ✓.
20	B	Monopoly with identical cost curves vs perfect competition: lower output, higher price, triangle of deadweight loss in consumer + producer surplus.
21	C	Deadweight loss triangle = the welfare loss from units that would have been traded without the tax but are no longer traded (between Q_1 and Q_2 , bounded by demand and supply curves).
22	B	Steeper SRAS (closer to LRAS) → larger price-level response to AD shifts, smaller output response. Inflation cost of stimulus rises as the output gap closes.

Question	Answer	Brief explanation
23	B	Classic weakness of the kinked demand curve: it explains price stability AT a given price, but not how that price was initially set.
24	B	Adverse supply shock (e.g. higher energy prices) shifts the SRPC outward — higher inflation at every unemployment rate (stagflation).
25	B	Excess capacity — the classic efficiency loss of monopolistic competition. Firms operate at output below the minimum of AC (productive inefficiency) and $P > MC$ (allocative inefficiency).
26	A	J-curve short-run worsening: elasticities are low in the short run (contracts fixed, habits slow to change), so import bills rise immediately while export volume response is delayed.
27	B	Ad valorem = percentage. The absolute tax amount rises with the price of the good. Specific tax is flat £ per unit regardless of price.
28	A	Gini = (area between Lorenz and equality) / (total area under equality diagonal). Standard definition.
29	A	In the inelastic region, TR rises with P. A profit-max monopolist would continue to raise price until reaching the elastic region — NEVER operates in the inelastic region.
30	B	Emissions permits set at social optimum quantity → market trading reallocates abatement to lowest-cost firms → efficient outcome at Qs.

Section B

Case study: Artificial intelligence, labour markets and inequality

Generic marking grid for the 10-mark data-response question

Level	Response	Max 10 marks
3	Good response that: • uses the data accurately and effectively to support the argument • identifies relevant economic concepts and applies them to the question • reaches a supported judgement using evidence from the extracts.	8–10 marks
2	Reasonable response that: • uses some data to support the argument but may lack precision • shows understanding of relevant economic concepts • reaches a judgement, though support may be limited.	4–7 marks
1	Weak response that: • makes limited reference to the data • shows limited understanding of relevant economic concepts • offers a judgement with little support.	1–3 marks

- 3 1** Using the data in the Extracts, examine the extent to which the rapid adoption of AI technologies has affected UK productivity and income inequality since 2020. **[10 marks]**

Relevant issues include:

- Evidence of rapid AI adoption (Extract A):
 - UK business AI adoption: 12% (2020) → 43% (2024) — more than tripled
 - AI-related job vacancies: 1.1% → 5.4% of total (nearly 5x increase)
- Evidence on productivity (Extract A):
 - Real GDP per hour worked rose from 100.0 (2020) to 102.8 (2024) — just +2.8% over four years, or roughly 0.7% per year
 - This is faster than the post-2008 average (0.4% p.a.) but not transformational
 - Link to AI is plausible but not yet firmly established — productivity boost could also reflect other factors
- Evidence on inequality (Extract A, C):
 - Top/bottom earnings ratio rose from 4.0 (2020) to 4.5 (2024) — an increase of 12.5%
 - AI vacancies command 50% wage premium over median (Extract C)
 - Regional inequality widening as AI jobs cluster in London/SE
- Caveats:
 - Correlation not causation: inequality was rising before AI; other factors contribute
 - Short observation window: full productivity effects may take years to materialise (Solow paradox: 'you can see the computer age everywhere but in the productivity statistics')
 - Distributional effect is clearer than aggregate productivity effect
- Concluding judgement:
 - Moderate evidence that AI has contributed to a slight productivity acceleration
 - Stronger evidence of an inequality-widening effect through the technology premium
 - The full impact is likely to unfold over a longer time horizon

Levels of response marking grid — 15-mark questions

The following generic grid should be used when marking all 15-mark questions in Section B (questions 0 9, 1 1 and 1 3).

Level of response	Response	Max 15 marks
3	A good response that: • is well organised and develops a selection of the key issues that are relevant to the question • shows sound knowledge and understanding of economic terminology, concepts and principles with few, if any, errors • includes good application of relevant economic principles to the given context and, where appropriate, good use of data to support the response • includes well-focused analysis with clear, logical chains of reasoning.	11–15 marks
2	A reasonable response that: • focuses on issues that are relevant to the question • shows satisfactory knowledge and understanding of economic terminology, concepts and principles but some weaknesses may be present • includes reasonable application of relevant economic principles to the given context and, where appropriate, some use of data to support the response • includes some reasonable analysis but which might not be adequately developed or becomes confused in places.	6–10 marks
1	A weak response that: • has identified one or more relevant issues • has some limited knowledge and understanding of economic terminology, concepts and principles but some errors are likely • has very limited application of relevant economic principles to the given context and/or data to the question • might have some limited analysis but it may lack focus and/or become confused.	1–5 marks

- 3 2** Using Extracts A, B and C and your knowledge of economics, explain why the adoption of AI technologies may increase both productivity and income inequality in the UK economy. **[15 marks]**

Relevant issues include:

- Productivity channel (AI RAISING productivity):
 - AI acts as capital-deepening — complements human labour, raising output per hour
 - R&D; and technology spillovers: AI developed in one firm diffuses across the economy
 - Reallocation effects: AI adoption encourages reallocation of labour from lower-productivity to higher-productivity firms
 - Extract A evidence: output per hour rose 2.8% over 2020-24, faster than post-2008 trend
 - Theoretically: rightward shift in LRAS (diagrammatic treatment expected)
- Inequality channel (AI RAISING inequality):
 - Skill-biased technical change: AI complements skilled workers (higher MRPL), substitutes for routine cognitive workers (lower MRPL)
 - 'Technology premium': AI-related vacancies pay 50% above median (Extract C)
 - Capital vs labour share: if AI substitutes for labour, capital owners capture more of the productivity gains (Acemoglu 2024)
 - Distribution of AI adoption: concentrated in larger firms, urban areas, educated workers — exacerbating existing gaps
 - Regional inequality: London/SE captures most AI jobs; northern regions benefit less (Extract C)
 - Labour share of national income may fall if AI is strongly substituting
- Linking the two:
 - The SAME adoption process that raises productivity (substitution of AI for certain labour) raises inequality (displaced workers, capital gains)

- • In theory, productivity gains could fund redistribution — whether this happens is a policy choice
- • Historical parallel: industrial revolution raised both productivity and inequality for decades before distribution improved
- Key economic concepts: skill-biased technical change; elasticity of substitution between capital and labour; labour-share dynamics; marginal productivity theory of wages

Use the levels mark scheme for 15-mark questions to award candidates marks.

Generic marking grid — 25-mark recommendation question

Level of response	Response	Max 25 marks
5	Sound, focused analysis and well-supported evaluation that: • is well organised, showing sound knowledge and understanding of economic terminology, concepts and principles with few, if any, errors • includes good application of relevant economic principles to the given context and, where appropriate, good use of data to support the response • includes well-focused analysis with clear, logical chains of reasoning • includes supported evaluation throughout the response and in a final conclusion.	21–25 marks
4	Sound, focused analysis and some supported evaluation that: • is well organised, showing sound knowledge and understanding of economic terminology, concepts and principles with few, if any, errors • includes some good application of relevant economic principles to the given context and, where appropriate, some good use of data to support the response • includes some well-focused analysis with clear, logical chains of reasoning • includes some reasonable, supported evaluation.	16–20 marks
3	Some reasonable analysis but generally unsupported evaluation that: • focuses on issues that are relevant to the question, showing satisfactory knowledge and understanding of economic terminology, concepts and principles but some weaknesses may be present • includes reasonable application of relevant economic principles to the given context and, where appropriate, some use of data to support the response • includes some reasonable analysis but which might not be adequately developed or becomes confused in places • includes fairly superficial evaluation; there is likely to be some attempt to make relevant judgements but these aren't well-supported by arguments and/or data.	11–15 marks
2	A fairly weak response with some understanding that: • includes some limited knowledge and understanding of economic terminology, concepts and principles but some errors are likely • includes some limited application of relevant economic principles to the given context and/or data to the question • includes some limited analysis but it may lack focus and/or become confused • includes some evaluation which is weak and unsupported.	6–10 marks
1	A very weak response that: • includes little relevant knowledge and understanding of economic terminology, concepts and principles • includes application to the given context which is, at best, very weak • includes attempted analysis which is weak and unsupported.	1–5 marks

3 3 Considering the information in Extract D and the original evidence in Extracts A, B and C, do you recommend that the UK government should adopt active interventionist policies (such as labour-protective regulation, a universal basic income or targeted AI taxation) rather than relying on market adjustment to manage the economic effects of AI? Justify your recommendation.

[25 marks]

Areas for discussion include:

- Case FOR interventionist policies:
 - Rising income inequality (Extract A, C): top/bottom ratio 4.0 → 4.5
 - Regional inequality (Extract C): AI jobs concentrated in London/SE
 - Market failures specific to AI:
 - Externalities (social cost of displacement, mental-health effects)
 - Concentration of market power among AI developers (positive feedback loops)
 - Information asymmetry (workers unaware of AI risks to their jobs)
 - Speed of change: AI advances too fast for labour-market adjustment alone
 - Equity / distributional concerns: labour share may fall if AI substitutes strongly
 - Specific policy instruments:
 - Retraining / active labour market programmes
 - Strengthened labour rights and collective bargaining
 - UBI as insurance against AI displacement
 - Targeted taxation of AI-related profits / automation
 - Competition policy to prevent dominance of a few AI firms
- Case AGAINST interventionist policies (case FOR market adjustment):
 - Historical track record: technology transitions have created more jobs than destroyed (printing press, factories, computers)
 - Supply-side flexibility: markets reallocate labour efficiently when regulation is not restrictive
 - Government failure risks:
 - 'Robot tax' slows AI adoption → UK loses competitiveness to other economies
 - Regulatory capture: big AI firms lobbying for entrenchment
 - UBI may reduce labour supply
 - Labour-protective regulation may entrench incumbents and hurt job creation
 - Supply-side reforms (education, skills) are the historically effective response
 - Alternative / complementary approaches:
 - Hybrid: light-touch regulation + heavy investment in retraining
 - Sector-specific: strong labour protection in AI-exposed sectors only
 - Output-contingent: deploy interventions if employment deteriorates, not preemptively
 - International coordination: uniform AI taxation to avoid jurisdiction arbitrage
- Evaluation:
 - The binding constraint is DELIVERY, not just policy direction
 - Interventions should be carefully targeted at the specific market failures identified
 - The rapid pace of change argues for adaptive, not fixed, policy design
 - Distributional concerns are strongest when labour share is shrinking
 - Historical parallels: post-war welfare state + education expansion handled earlier technology transitions — the question is whether today's transition is qualitatively different

Indicative judgement:

- A reasoned recommendation — YES, NO, or HYBRID — can reach Level 5 if well-supported
- Strongest candidates will distinguish between different types of intervention: some (retraining, competition policy) are clearly justified by market failures, while others (robot tax, blanket UBI) are more controversial
- Candidates who recognise the distributional vs allocative trade-offs in AI policy deserve reward

- Those who simply summarise both cases without clear judgement should be placed in Level 3
- Candidates who engage critically with Acemoglu's argument about labour share (Extract C) and with historical parallels to previous technology transitions should be rewarded highly

Use the levels mark scheme for 25-mark questions to award candidates marks. Effective use of extracts and relevant diagrams should be rewarded.

END OF MARK SCHEME